



COURSE SYLLABUS

CSC15106 – Knowledge Engineering Seminars

1. GENERAL INFORMATION

Course name:	Knowledge Engineering Seminars
Course name (in Vietnamese):	Seminar Công nghệ Tri thức
Course ID:	CSC15106
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Introduction to Artificial Intelligence
Prior-course:	Machine Learning
Instructors:	Nguyen Dinh Thuc

2. COURSE DESCRIPTION

The course is designed to provide students with an overview of the major research directions in the Department of Knowledge Engineering (KE). It consists of a series of seminars about various research topics in the fields of artificial intelligence (AI) and machine learning (ML), spoken language processing (SLP), and information security (IS). Each topic will be presented by experts in the fields, from advanced research to practical applications. In addition, the course will further provide students with the basic research skills, including proposal writing and research results representation.

Students of the Knowledge Engineering Department are encouraged to take this course to discover their research interests, as well as to get ready for their thesis.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Understand the basic concepts of AI and the main research directions of the Department of Knowledge Engineering.	1.7.1 1.7.2
G2	Understand the main research topics and solutions in spoken language processing.	1.7.2
G3	Understand the fundamentals of cryptography and its applications in security systems and blockchains.	1.7.2
G4	Write a research proposal and present a research result.	4.1

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Understand and discuss the main research directions in the Knowledge Engineering Department.	I
G1.2	Understand the basic concepts and knowledge of AI and ML.	I, T
G2.1	Understand the major research topics and solutions in natural language processing.	I, T
G2.2	Understand the major research topics and solutions in automatic speech recognition.	I, T
G3.1	Understand the fundamentals of cryptography.	I, T
G3.2	Know some typical applications of crypto and blockchain.	I, T
G4.1	Survey a research topic, formulate a problem, and write it into a research proposal.	T, U
G4.2	Design and represent a research result.	T, U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Course introduction	G1.1	Lecturing, Q&A

2	Seminar on machine learning	G1.2	Lecturing Demonstration, Q&A
3	Seminar on natural language processing 1	G1.2	Lecturing, Q&A
4	Seminar on natural language processing 2	G1.2	Lecturing Demonstration, Q&A
5	Seminar on speech technology	G2.1	Lecturing, Q&A
6	Seminar on information security 1	G2.2	Lecturing, Q&A
7	Seminar on information security 2	G3.1, G3.2	Lecturing, Q&A
8	Seminar on literature reviewing	G4.1	Lecturing, Q&A, discussion
9	Seminar on technical presentation	G4.2	Lecturing, Q&A, discussion
10	Review	G1.1, G1.2	Lecturing Q&A, Discussion Project submitted

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			40%
A11	Assignment on machine learning and deep learning	Build deep learning for a problem in pattern recognition	G1.1, G1.2,	10%

A12	Assignment on natural language processing	Build solution for problems in natural language problems	G1.1, G1.2, G2.1, G2.2	10%
A13	Assignment on information security	Implement algorithms in information security	G3.1, G3.2	10%
A14	Report about a research topic in information security and NLP	Follow a topic in information security and NLP during the seminar and write a report	G1.1, G1.2, G2.1, G2.2, G3.1, G3.2	10%
A2	Projects			40%
	Study a problem and solutions for it	• Survey a research topic of interest • Study a problem with clear objectives and solutions for it • Implement solutions and applications • Present solutions and demo applications	G4.1, G4.2	40%
A3	Exams			20%
	Final exam	Q&A on the final project	G4.1, G4.2	20%

7. RESOURCES

1. Textbooks

- **Artificial Intelligence**, Earl B. Hunt, *Academic Press Series in Cognition and Perception*, London, 1975
- **Handbook of Natural Language Processing**, Nitin Indurkha, Fred J. Damerau, *Chapman and Hall/CRC*, New York, 2010
- **Deep Learning**, Ian J. Goodfellow, Yoshua Bengio and Aaron Courville, *MIT Press*, 2016

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.